

Research Article

Analysis of Service-Oriented Architecture and Scrum Software Development Approach for IIoT

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Flexibility and change adoption are key attributes for service-oriented architecture (SOA) and agile software development processes. Although the notion of agility is quite visible on both sides, still the integration of the two diverse concepts (architectural framework and development process) should be well thought of before employing them for a software development project. For this purpose, this study is designed to analyze the two diverse software architectural framework and development approaches, that is, SOA and Scrum process model, respectively, and their integrated environment in software project development setup perspective for Industrial Internet of Things (IIoT). This study also analyzes commonalities among Scrum process model and SOA architectural framework to identify compatibility between Scrum and SOA so that the Scrum process can be constructively used for SOA based projects. This study also examines the proper design and setup of Scrum process suitable for large-scale SOA based projects. For this purpose, an SOA based research and development project is selected as a case study using Scrum as the software development process. The project development and deployment perspective include eight core modules that constitute the overall project framework.

1. Introduction

In the present era of dynamic business environment, flexibility to welcome change and adapting to it efficiently and cost effectively are pertinent to the success of any business organization. Service-oriented architecture (SOA) is an architectural approach to design, develop, manage, and deploy a software application and infrastructure in which all the applications are structured into business services that are network accessible and executable [1]. Flexibility and change adoption are key attributes for SOA and agile software development processes. Although the notion of agility is quite visible on both sides, still the integration of the two diverse concepts (architectural framework and development process) should be well thought of before employing them

for a software development project [2]. Therefore, the use of an appropriate agile process for the SOA based application development, to adopt major requirements modifications and changes even during application building along with the conservation of software superiority and quality, is essential [3]. Agile processes (based on agile manifesto [4]) tend to focus on iterations and client suggestions to improve performance and allows for the predictability of varying requirements. Agile software development (ASD) is the development process through which a system is developed efficiently and rapidly by means of regular, frequent, and complete releases permitting the participants and stakeholders to get their hands on the application [5, 6]. The application and process are reviewed and tested through agile retrospective meetings. In this way, a prototype is

developed into useful progressive and creative system by means of an iterative and incremental process, whereby feedback is given by the stakeholders, based upon the rapid successive releases of the software.

Iteration stresses on the delivery of working software that provides grade and value to customers as well as to project. It also produces other concerned artifacts which are valuable to both customer and project. The aim of agile process is to provide workable deliverables in a dynamic way. Proponents believe that in an agile process the main concern is to provide deliverables and working products in a dynamic way without emphasizing on design models and documentation perspective [7, 8]. Scrum is one of the agile methodologies, which is a standard way of introducing agility due to its flexibility and straightforwardness [9]. It is a popular agile management method in industry. Agile development of applications in an enterprise surrounding can be challenging because of the compound nature of team members and their environment [10]. Scrum process facilitates discovering better ways of developing software by promoting individuals as well as teams [11]. Scrum divides the development process into iterations, called sprints, where a sprint stresses on the delivery of working product that provides value to both the project and customer [12]. Scrum, as the most used agile process, highlights empirical feedback, team self-management, and struggle to build properly tested product increments within short iterations [13]. The product owner, Scrum team, and Scrum master are the main three roles in Scrum. The responsibilities of the traditional project manager role are split among these three Scrum roles [14]. Scrum master is the one who maintains the process of the team, leads Scrum meetings, and makes sure that the process esteems all rules and principles of Scrum. Scrum team is a cross-functional team (having different expertise in different areas) involving developers, designers, testers, and analysts. This team is responsible for the project as a whole and is also involved in the development. Product owner is the one who represents the interest of end users and others interested in the product parties.

Scrum is based on regular meetings in which Scrum masters, product owner(s), the developers, and third parties discuss different issues concerning the development process. These meetings are the following: sprint planning meeting, daily Scrum meeting, sprint review meeting, and sprint retrospective meeting. Sprint planning meeting is held at the beginning of every sprint cycle. During this meeting, the product owner and the Scrum team define sprint goal which the sprint will attempt to achieve [15]. The success of the sprint will later be assessed during the sprint review meeting against the sprint goal. In sprint review meeting at the end of each sprint, the team demonstrates completed functionality and shows what they have accomplished during the sprint. In sprint retrospective meeting, team members review the way the team works and interacts, behavioral aspects, and improving technical skills, so that the subsequent sprint is faster, and so forth.

The Scrum process is a black box approach, where the team and processes are built and discovered dynamically during the project working, whereas nonagile approaches

such as waterfall model are signified as a white box nature of process model from the management viewpoint [14]. Planning and postmortem phases are important because these identify goals and outputs produced in a sprint. Planning is vital for overcoming any blockage in any phase or process during development. In agile terminology, the postmortem meeting known as retrospective is important for looking at the processes and creating knowledge base through best practices. Figure 1 shows the analogy of white box and black box nature of processes models.

Nonagile development methodologies are heavy-weight processes for developing software. These methodologies are based on a successive sequence of steps, for example, requirements explanation, solution structure, testing, and deployment. Traditional methodologies impose heavy documentation and define a fixed set of requirements at the start of a project. There are various traditional methodologies such as waterfall, spiral model, win-win spiral model, and unified process. The heavy-weight processes require detailed upfront plan, heavy documentation, and broad upfront design. Still traditional methodologies have their utility for large-scale projects that have predefined and considerably fixed set of requirements, although practitioners have also started to adopt agile approaches for these projects [16].

SOA is an architectural framework and approach to design, develop, manage, and deploy a software application. It is a software infrastructure in which all applications are structured into business logic called services that are network executable and accessible. In other words, SOA agrees to integration of applications, users, and existing system into a flexible architecture that can easily accommodate changes when they are needed in a system [13]. SOA is regarded as one of the best approaches for distributed application development. SOA allows reusing the functionality of existing systems rather than building again from scratch. This feature of reusability in SOA based applications maximizes economic benefits for the organizations [14]. Each service in SOA performs autonomously but is not isolated from the whole. Each service encapsulates a specific logic in the problem domain. The other features of SOA are loose coupling, service contract, autonomy, abstraction, discoverability, and statelessness [17].

Although SOA and agile approaches are generally viewed with related concerns, still there is no clear definition of organization and setup of both approaches in a single environment for IIoT. Truly little information is provided as to what will be the impact of this integrated implementation on the important factors such as productivity, quality, agility, and innovativeness.

Based on the above claims and current market needs, this study analyzes commonalities among Scrum process model and SOA architectural framework to identify compatibility between Scrum and SOA so that the Scrum process can be constructively used for SOA based projects. This study also examines appropriate design and setup of Scrum process suitable for large-scale SOA based projects. For this purpose, an SOA based research and development project is selected as a case study using Scrum as the software development

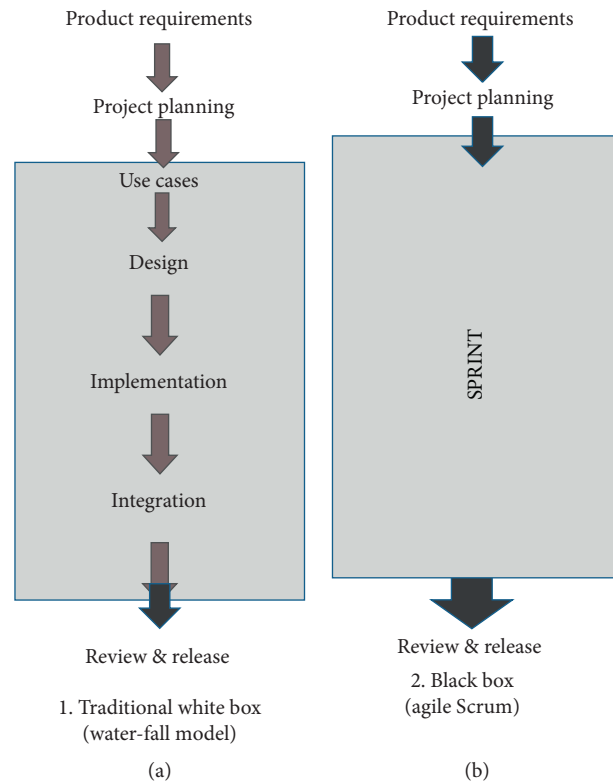


FIGURE 1: Analogy of white box and black box processes.

process. The project development and deployment perspective includes eight core modules that constitute the overall project framework.

This study is organized in seven sections. The current section introduces major terminologies used in this study. The second section presents analysis of existing research in Scrum and SOA approach for an in-depth understanding. The third section is dedicated to studying design and setup, analysis of SOA and Scrum, and integration of Scrum and SOA. The fourth section is about the integration analysis; the fifth section is the discussion about SOA and Scrum integration, while the conclusion and future work of overall study are presented in the sixth and seventh sections, respectively.

2. Literature Study

Agile processes are planned to upkeep early and fast development of software applications. This is made possible by working in iterations [9]. Iterations stress on the delivery of working software that provides value to customers as well as to the project. An iteration is a short development cycle that produces working software and other artifacts which are valuable to both customer and the project [9]. Main concern of agile processes is to provide deliverables and working products in a dynamic way without emphasizing on design models and documentation perspective [7].

Experts of traditional development process conclude that when documentation and analysis process is neglected, it leads to corporate memory loss, especially when complex

and large-scale software are developed. But the experts of agile confirm that, by following agility for software development global software delivery, it has led to better outcomes in gaining concerned objectives [18, 19]. It is clear that organizations will have to face some sort of challenges in coordinating and integrating agility and global service delivery, but also it makes organizations able to deliver quick and error-free software that meets concerned business requirements [20].

Agile software development methodologies are introduced to provide answers to the concerned business community which asks for light-weight, more rapid, and more flexible software development methods [21]. Therefore, agile development methodologies are dedicated for agility, quickness, and eagerness for development [22]. According to Oxford dictionary, agile processes are devoting the importance of being agile, willingness for motion, nimbleness, activity, and dexterity in motion [12]. Some of the agile methodologies like extreme programming and Scrum have acknowledged that customer satisfaction could be achieved through the lightness of concerned processes [12].

Scrum is one of the most used agile management methods. According to last three agile adoption surveys, Scrum highlights empirical feedback, team self-management, and struggling to build thoroughly tested product increments within short iterations [23, 24]. Scrum is an agile methodology which is the most standard way of introducing agility due to its flexibility and straightforwardness and a popular management agile method in industry. Agile software development process facilitates better ways of

developing software by promoting individual work as well as teamwork [11]. Agile processes are planned to maintain early and fast development of software application. This is made possible by dividing the development process into sprints, where sprint stresses on the delivery of working product that provides value to both the project and customer [25]. Agile development of applications in an enterprise surrounding can be challenging because of the compound nature team members and their environments [11].

SOA agrees to the integration of applications, users, and existing system into a flexible architecture that can easily accommodate changes when they are needed in a system [14]. SOA is regarded as one of the best approaches for distributed application development [26]. SOA allows reusing the functionality of existing systems rather than building again from scratch [27]. This feature of reusability in the SOA based applications maximizes economic benefits for the organization. Each service in SOA performs autonomously but is not isolated from the whole system. Each service encapsulates a specific logic in the problem domain. The main features of SOA are reusability, loose coupling, service contract, autonomy, abstraction, discoverability, and statelessness [28].

Researchers and professionals have a mixed opinion about similarity and compatibility of the two approaches. Critics emphasize differences among SOA and agile approaches, arguing that SOA and agile are standing at different development direction: SOA is architecture and agile is a methodology [24, 26]; SOA works in a top-down manner, while agile is inherently a bottom-up approach [29, 30]. SOA is an architectural framework and follows a set of principles, whereas agile is a process model and works at practice level [31]. Based on this view, advocates of SOA and agile integration suggest that one way of combining them is to use SOA framework and guiding principles of agile practices for service development [32, 33]. Some researchers also claim that SOA based systems are developed and deployed differently from traditional Systems [21]. Also, there are many challenges like stakeholders' involvement, business and IT alignment, and reuse of assets. To overcome such type of problems, agility and service orientation are better integrated [34, 35]. It is notable that Scrum and SOA share similar concerns, such as responsiveness to change, new ways of working, flexibility, and business understanding [36].

SOA and Scrum assessment and specification need attention of technology factors along with their relationships to IT related business organization [37] and to individual information consumers. Other requirements such as flexibility and ease of use are defined in terms of such type of system features that can be measured and designed [38, 39].

3. Design of SOA and Scrum Based Environment for IIoT

Scrum as a development process and SOA as an architecture framework deal with similar concerns, for example, flexibility and quick response to change with profound business orientation. Even though this inherent operational similarity

is quite obvious, still not enough research has been made in analyzing the most optimal solution for the integration of these two approaches in order to receive the combined benefits [40, 41]. Also, there is a need to investigate the impact of employing SOA with regard to improving productivity and efficient utilization of time and resources using Scrum methodology.

To investigate this context, a case study has been selected, where Scrum process model is used as a development methodology for the development of an SOA based software system. The SOA based project is named M4S (Mineral resource, Mapping, Modeling, and Management System). The project development and deployment perspective includes eight core modules that constitute the overall project framework. The system is developed following the systematic steps which are discussed in detail in the following paragraphs.

This study is evaluated through identifying, determining, and using one of the best combinations in software engineering, which is designed based on SOA, using highly productive open-source frameworks and tools, development through agile development methodologies, and deployment as a combination of client and cloud applications.

Scrum has been used as an agile process for continuous development and improvement to following changing requirement to refine working prototypes in iterations [42, 43]. The application is designed using SOA to ensure that the case study system modules can be used as independent subsystems through service interface. SOA facilitates exploiting the characteristics and capabilities of both Scrum and SOA for integration, customization, and binding of the desired designed frameworks and tools, the Mineral resource, Mapping, Modeling, and Management System (M4S); ultimately the deployment phase of this system has infrastructure having properties of interactive communication, scalability, and dependability of services.

These deployment requirements are achieved through cloud infrastructure and distributed computing. Modules of the developed application are designed to act as interoperable services and can be used independently and recombined in other modules resulting in an integration of multiple coherent modules. Data banks for mineral, resources, and application to manage those data banks are specifically designed for M4S system. These databanks, libraries, and applications are designed to be used as open-source utilities.

Scrum methodology is used to ensure having a transparent process of analysis and design of M4S and engagement of stakeholders and also ensures a dynamic interaction among the different components of the project. The use of agility and Open-Source System (OSS) principles for the development and project management has permitted targeting a diverse community of stakeholders through multilateral liaison of academia, government, and industry. A collaborative work of these diverse actors played a prime role in M4S development, along with a minor role of individual contributors.

As discussed in ICT R&D project named M4S [<https://www.openm4s.org>], the core framework of M4S utilizing open-source libraries, services, servers, applications, database management system, and an open-source Enterprise resource

planning (ERP) is shown in Figure 2. Client-server architecture and browser-based clients are based on the principles of interoperability and open standards. The figure also shows components and information flow within proposed M4S.

Scrum-driven agile software development methodology is used for managing and organizing the development of modules [15]. Scrum used in the context of this case study along with specified roles and team organization is given in the following section: The process setup is decided according to the specification of agile Scrum. The following is a brief description of roles as these are present and work in a Scrum process environment.

- (a) Scrum master is the one who leads Scrum meetings, maintains the processes of team, and makes sure that the process respects all principles of Scrum [44, 45].
- (b) Scrum coach also facilitates meetings, helps the team to reach consensus, and takes care of quality aspects of the development environment and team.
- (c) Scrum team is a cross-functional team consisting of developers, testers, architects, and analysts. This team is responsible for project results as a whole and is fully involved in development. The team takes care of developing features according to quality and functional requirements [46].
- (d) Product owner is a person who represents the interests of end users and the stockholders. The product owner takes care of quality aspects related to user requirements. A typical sprint cycle is presented in Figure 3.

3.1. Team Organization of M4S Process. Team organization of M4S process is shown in Table 1 for the project development. Product owners provide the roadmap, business case, release plans, and architectural direction. Scrum masters have the knowledge on the architectural framework, usage context of the requirements, and features of M4S. They performed the business analysis and quality assurance tasks, apart from managing the sprint cycles.

Key roles and responsibilities of each member in the project are as follows:

PD (project director) was responsible for planning, monitoring, and managing project activities.

Co-PD (coproject director) was responsible for monitoring project activities, establishing liaison, and coordination with government and industrial partners.

Software and computer systems experts were responsible for directing software architecture design and quality assurance.

Engineering management expert was responsible for directing project management activities and quality assurance.

Team leads were responsible for scheduling and execution of design, development, and deployment tasks.

Developer teams were responsible for development, testing, verification, and documentation of modules.

Coproject director was responsible for maintaining liaison and coordination with government and industrial partners.

Mining and software experts (part time) were responsible for testing and validation of M4S and its modules.

Mentors (student supervisors) were responsible for supervision and guidance to UG and PG students, working on M4S related projects.

Part-time developers were responsible for development, testing, and verification of modules.

Internees were working on various features of M4S as their research projects and developed and documented the assigned projects.

Work breakdown structure (WBS), along with the responsible personnel for each block of activity, of M4S project is shown in Figure 4.

The following are the details of the Scrum process and its different features which are applied in the project.

(a) User stories

A user story captures what the user wants to achieve through the system. This objective is transformed into a user story from the customer's point of view [47]. In accordance with the user stories, acceptance test card is prepared by the product owners (project director and codirector) and the developers implement those stories. The acceptance test card comprises a set of test cases against which the implemented user stories are tested. A user story will be considered complete only if the acceptance tests are satisfactory. Once ready, the user stories are turned into system features [17]. A feature means "work to be done" or a piece of functionality to be implemented in the system.

In M4S project, individual features are decomposed into smaller tasks. Then story cards are designed for each individual task. The completion time for a task was recommended to be from five to fifteen working days, and a working day was normally consisting of eight hours. Normally one or two team members are working on only one story card at a time. Each story or task is assigned priority according to its importance. Team members select stories on their priority base to finish.

Figure 5 depicts two stories that are finished before the selection of next story.

(b) Sprint

According to Scrum process, development is organized into two- or four-week-long periods called sprints. A sprint is a time-boxed development iteration within which selected tasks are implemented [37]. Tasks are assigned to sprints during a sprint planning session at the beginning of each sprint based on the status of the backlog. A one-week sprint (also called sprint 0) is recommended before starting with the M4S project development. This allows the developers to get acquainted with the development

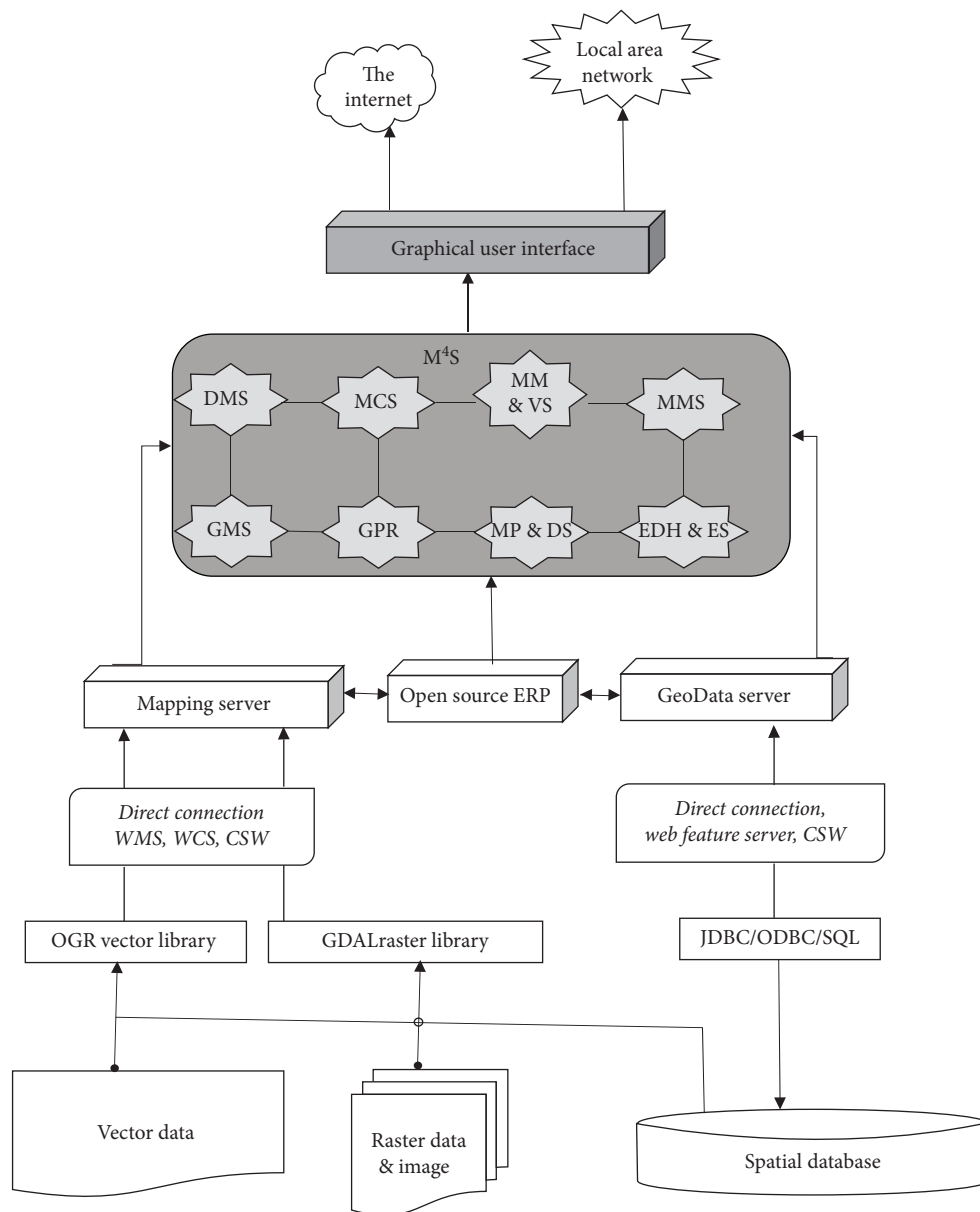


FIGURE 2: Proposed frameworks of M4S [<https://www.openm4s.org>].

environment, technological solutions, and the existing components that are reused.

The first module of M4S is completed in seven sprints. The first sprint consists of total nine stories, in which seven stories were completed within sprint time box, while the remaining two stories were included in the second sprint to be completed next. Each sprint is designed in Microsoft Excel sheet, which contains all the information about each story, team members who worked on a particular story, time required for a particular story to be completed, and so forth. Figure 6 depicts the first sprint as a sample as follows.

Scrum methodology is based on regular meetings where product owner, Scrum team, third parties, and

Scrum master discuss different issues faced/accrued during development time. The meetings are held for the purpose of ensuring smooth running of the project and preserving/maintaining the overall quality phases of the developed system. Scrum process model employs five types of meetings which are daily Scrum meeting, sprint retrospective meeting, sprint review meeting, sprint planning meeting, and sprint release meeting. These five meetings are also held and maintained during the M4S implementation in different situations. These meetings are discussed one by one in the following subsection.

3.1.1. Sprint Planning Meeting. A total of seven sprint meetings have been held to plan the work to be finished in a sprint. Normally a meeting is held after every two weeks.

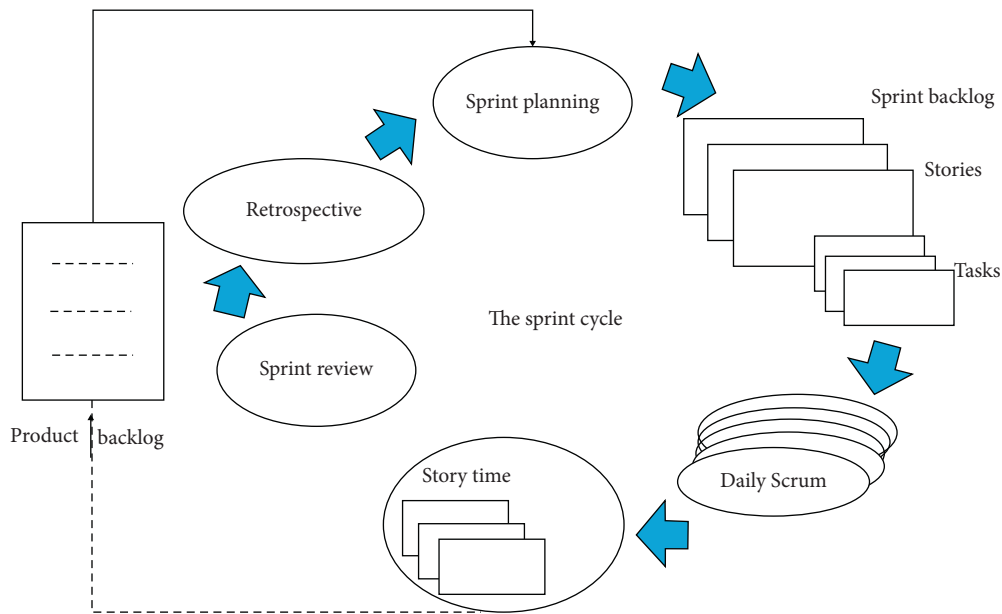


FIGURE 3: Typical sprint cycle.

TABLE 1: Scrum teams for M4S.

Product owner	Product director, coproject director
Scrum master	Team leader
Scrum team	Senior developers Internees (graduate & undergraduate students) Developers and experts (S/W and CS experts, engineering management experts, mining and software engineers, mentors, and part-time developers) integrated in the team for special task like testing and quality assurance Stakeholders and coordinators from industry, government institutions, and academia (analysts)

During this meeting, Scrum team and the product owner (M4S project director & codirector) define the sprint goals. The achievement of the sprint is later evaluated in the sprint review meeting. Product owner works to prioritize the story cards, to describe the highest priority story points to the Scrum team. Finally, Scrum team needs to prepare the sprint plan and sprint backlog that detail the time it takes to do the work [48].

3.1.2. Daily Scrum. A daily Scrum meeting is held, the main purpose of which is to make sure that application's features are being implemented in order according to their planned agenda. Normally this meeting takes place during each sprint in the starting time of work and lasts for approximately 20 to 35 minutes. The meeting is scheduled at exact regular time. During this meeting, each member of the team will answer the succeeding three questions:

- (i) What and how much work have you been doing since yesterday?
- (ii) What are the plans for today work?
- (iii) What are the impediments that would prevent accomplishing the goal?

If any issue occurred, then it is the responsibility of the team leader (formally called Scrum team) to simplify and resolve these complications. Unsolved issues are rediscussed in another meeting. This meeting is held in the same place, at the same time, and for the same duration.

3.1.3. Sprint Review Meeting. This meeting is held after each sprint. In this meeting, the team members show what they have done and achieved during the completed sprint. Normally, this meeting takes place in the form of demonstration of the recently developed features. In M4S project, Scrum team has finished at least 95% of product backlog story points fetched to the sprint, and the team achieved most of the sprint goals. In each module of the M4S project, 98% of the features that meet the "done" criteria are accepted and marked as complete. The remaining 2% of the features that are not complete in the specified sprint are rescheduled for future sprints. The project director (product owner) is satisfied by the 98% completion; then the features are closed, and next backlog is updated to include the features that need further development.

- (a) The completed stories for each sprint are accepted by M4S project director and coproject director (product owners).

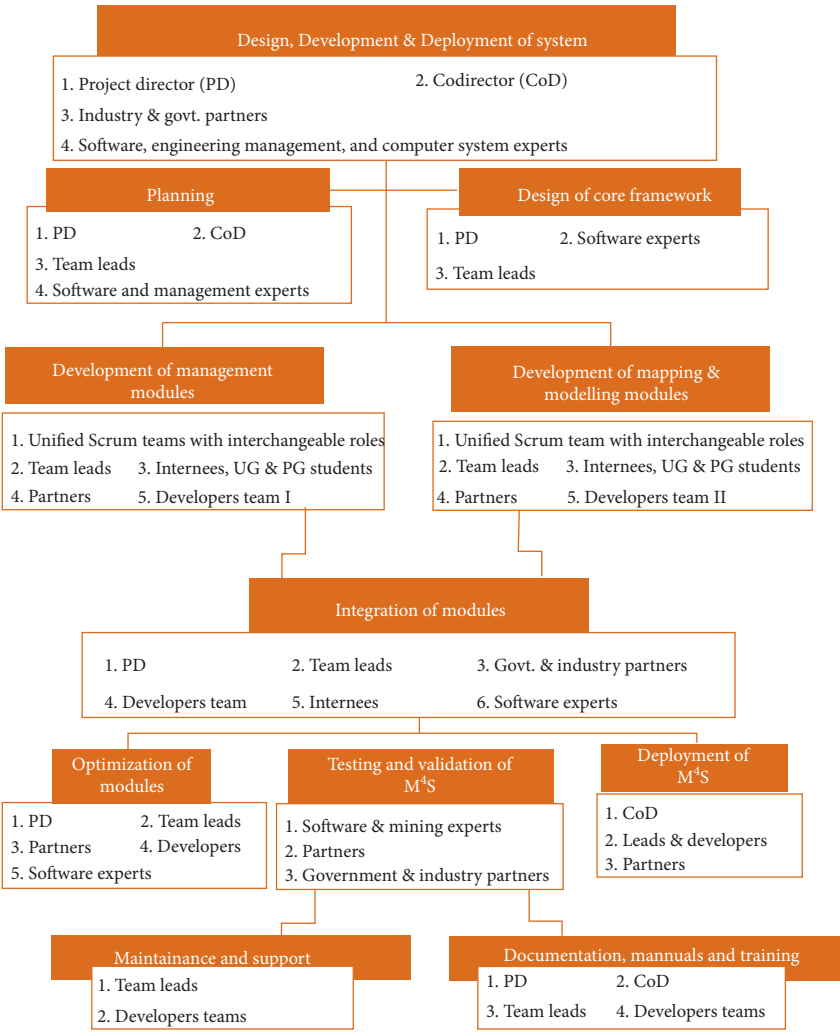


FIGURE 4: Work breakdown structure and resource personnel for M4S project [https://www.openm4s.org].

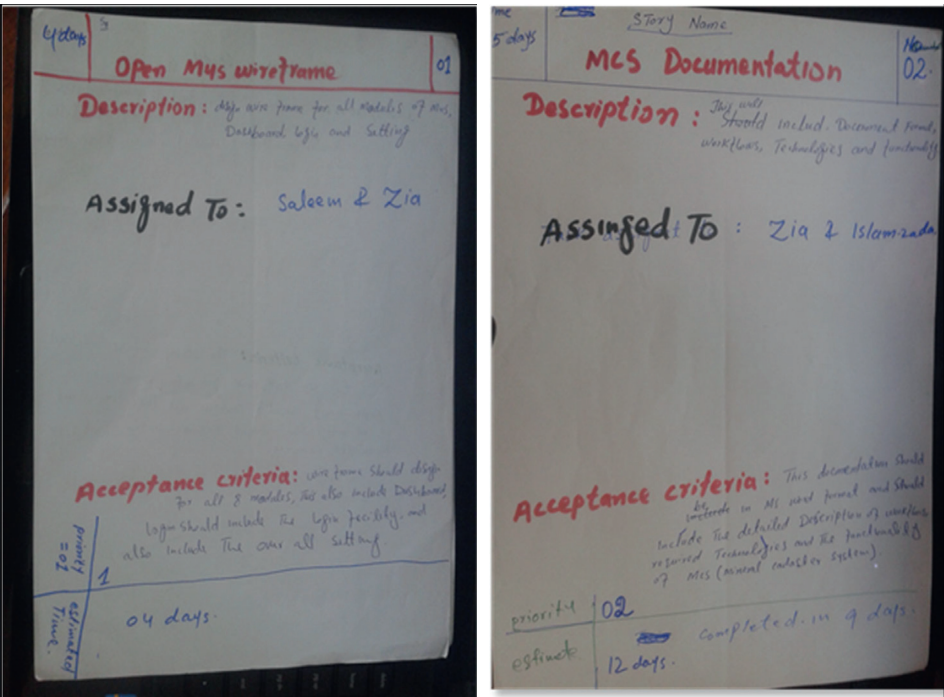


FIGURE 5: Story cards used in the project.

	A	B	C	D	E	F	G
1	M4S Sprint Planning						
2	Sprint #	Sprint:1					
3	Sprint Name	Wireframe & Documentation					
4	Sprint Goal	To build a base for Open M4S, Complete Documentation					
5	Team Member	8	Anwar,Qasim,Zia,Nahyan,Saleem,Zahid,Islam,Iftihar				
6	Sprint Start Date	3/3/2014					
7	Sprint Duration	12	Days				
8	End Date	17/03/2014					
9	Story Name	Tasks Description		Assigned To	Start Date	End Date	Status
10	MCS Documentation	Document Format, Workflows, Technologies, Functionality		Saleem	3/3/2014	17/03/2014	Done
11	Open M4S Wireframe	Wire frame for all modules, Dashboard, login and Settings		Zia	3/3/2014	17/03/2014	Done
12	GeoNetwork Study	Study ,Installation, Knowledge Sharing		Nahyan	3/3/2014	17/03/2014	Done
13	Open Geo	Study ,Installation, Knowledge Sharing		Zia	3/3/2014	17/03/2014	Done
14	Mapping, GIS	Study, Example, Knowledge Sharing		Iftihar	3/3/2014	17/03/2014	Done
15	Sola	Study ,Installation, Knowledge Sharing		Saleem	3/3/2014	17/03/2014	Done
16	SOA	Study ,Installation, Knowledge Sharing		Islam	3/3/2014	17/03/2014	In Progress
17	Wireframe(4 Modules)	Wire frame and documention		Qasim	3/3/2014	17/03/2014	Done
18	ERP Documentation	Study ,Installation, Knowledge Sharing		Zahid	3/3/2014	17/03/2014	In Progress
19							

FIGURE 6: First sprint of M4S project designed in MS excel.

- (b) The finished features are accepted by project director during the meeting.
- (c) All the acceptance tests run for the stories in concerned sprint.
- (d) The whole code has been passed through code reviewing process.
- (e) No critical bugs are found in the bug backlog.
- (f) All finished story points for release are accepted.
- (g) The products are tested properly and have not found any serious bugs.
- (h) A successful backup is made.
- (i) Most of the deployment documents are up to date.

3.1.4. Sprint Retrospective. In this meeting, M4S team members reread the approach team workings, interrelations, behavioral features, and refining methodological skills, so that the ensuing sprint is more rapid, and so forth. Scrum team and team leader (Scrum master) discussed three things: what went well, what did not go well, and what improvements can be considered in the next sprint. The overall performance of the team is evaluated (in the form of metrics) and improvement strategies (in the form of KPIs) are identified, which are adjusted in our previous article. It is suggested that team members should have to share technical skills and knowledge, should keep in continuous communication with product owner, and so on. This meeting is held at the end of each sprint and lasts approximately two to three hours.

3.1.5. Release Planning. A release planning meeting is a continuous and uninterrupted process of describing,

prioritizing, and splitting the system features in a release backlog. This involves the identification and commitment on the following:

- (a) Release goals.
- (b) Prioritized set of user stories (features to be developed).
- (c) Rough estimates of user stories.
- (d) Release dates.

The main input steps that are held during this meeting in M4S project are the following:

- (a) Team members estimate the user stories, i.e., make rough estimates of the relative size of the stories.
- (b) Establish velocity, i.e., determine how many story points are completed in each sprint; the details are given in our other article.
- (c) Compute forecast, i.e., date-based release is estimated to complete (velocity $\times$ number of sprints) story points. Functionality-based releases are estimated to complete in (total story points $\div$ velocity) sprints.

The highest priority stories, whose sum is no more than the number of story points (computed as mentioned above), are selected. For a functionality-based release, if the estimated completion date (computed above) is acceptable, all the stories are selected for the release.

4. SOA and Scrum Integration

This section analyzes the integration of Scrum, a software development process, and SOA, an architectural style. SOA

introduces an organized and well-defined software design, where change is supported through principles of agility along with the application of design patterns and standards to achieve the quality, efficiency, and productivity [9]. Service reusability and abstraction are applied for designing flexible and adaptable systems [16]. SOA uses agility at design level as new services are designed and old ones are evolved on the underlying architecture and not by changing the architecture itself. Scrum allows development in increments, facilitating faster feedback among customer(s) and development team. Scrum and SOA both support the corporate process that promotes setting up aligned businesses and development strategies.

The analysis of Scrum and SOA process in M4S project development environment showed that Scrum and SOA use similar principles to reach the joint goal of required software development. Even though SOA runs in an organized and well-controlled setup, it can benefit from agile Scrum at the same time, without any unsuited overlapping. A clear strategy to introduce change in increments should, however, be designed.

Use of Scrum for traditional applications or conventional architectures is also possible by describing requirements into a backlog and incremental deliverables for the customers. Nevertheless, it is expected to get increased developmental efforts and overall project cost [49].

To get proposed benefits and maximum output through an agile process, it should be kept in mind that the underlying software architecture and IT organization environment also support the process in terms of responsiveness and costs. Otherwise, the organization may end up getting only minimal benefit from the agile process.

As already discussed, SOA and Scrum are approaches that follow different directions. In the services development scenario, SOA approach follows from top-down approach (services are building in the top of SOA system), while Scrum follows bottom-up approach (starting from initial planning to prototype delivery) as using a process development methodology. The following question arises: how are these different approaches compatible with each other when employed together for a development process? Another question is, does SOA also follow the agility just like Scrum process? If the answer is yes, then how? Finally, how could these two approaches be integrated with each other to get benefits offered by both individually? This section refers to these questions and also discusses SOA and Scrum metrics having commonalities. The SOA and Scrum typical developmental framework is depicted in Figure 7.

Table 2 identifies most important metrics of SOA and Scrum, which can provide more value to business and to those whose aim is to use SOA and Scrum together in a software development venture.

Some of Scrum and SOA metrics are used for common purpose, which are integrated to be used for Scrum and SOA combination. These metrics are integrated from Scrum and SOA metrics which are used for the same purpose but using different terminologies. Through these metrics, development process can be streamlined when a

proper measure is taken, which will provide more business agility, flexibility, and compatibility for SOA and Scrum environment.

The “completed stories vs. planned stories” metric of Scrum and “new service created and used as percentage of total service” metric of SOA measure the production of product using ratio and percentage as a scale device. They both are used for service or work measurements but using different terminologies. Also, the three metrics which are team velocity, development time, and average development time to develop a service measure the team progress in terms of sprint and time required for a service which is to be completed in a particular sprint within time. Due to common goals of these metrics, these could be combined into team velocity metrics which will be considered as a metric for SOA and Scrum integration. For quality measurement, Scrum and SOA used separate metrics, but the purpose was the same; therefore these two metrics can be used as quality assurance metric for measuring the services’ quality using Scrum process model.

The team enthusiasm metric measures whether the team members are happy and work eagerly or not. When the team members are happy and work in a comfortable environment, they will work willingly and will follow the free-planned architecture policies and rules. They will work in a collaborative environment. The communication between team members will always be positive when they work eagerly due to happiness.

5. Discussion of Compatibility of Scrum and SOA

The main purpose of SOA and Scrum is to make the whole enterprise agile by using services as the building blocks for software applications [50]. Also software development through Scrum process model means to increase organization agility by bringing together Scrum practices that could increase communication, collaboration, and feedback [13, 14]. As already discussed, Scrum and SOA are generally viewed with similar concerns, but still there exist some diversities and incompatibility issues between the two approaches. The compatibility and commonalities of these two approaches are discussed in order to make a ground for the integration of both development approaches.

5.1. Compatibility of Scrum and SOA. It may sound confusing as to why there is a need to find similarities and compatibilities among Scrum and SOA. Scrum and SOA are two different paradigms as one is a process and the other is an architectural style. Still, it is logical and relevant to find compatibilities of both when used together, for example, to develop an SOA based system using Scrum as a software development methodology, while most SOA teams are subconsciously aware of the way of design and development of services. The whole focus revolves around service design policies. SOA’s nature inspires specific team makeup and style of communication within teams according to policies just like Scrum practices [15, 51]. It can be said that Scrum is

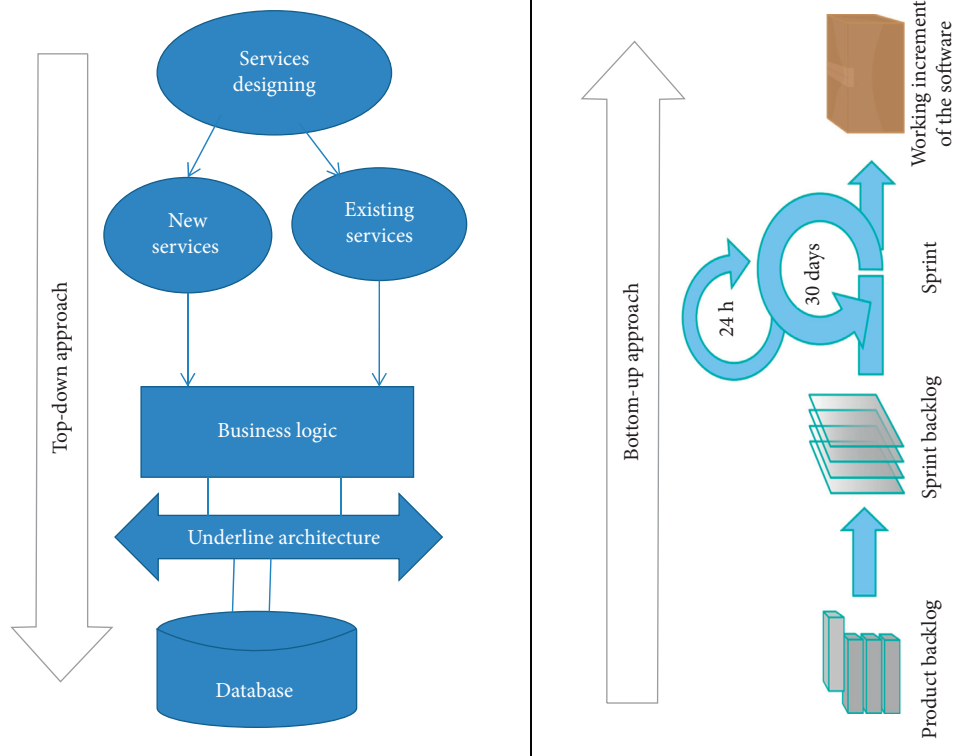


FIGURE 7: Typical developmental framework of SOA and Scrum.

like the human hands that work using gloves, while SOA is like that glove. It is understood that most of Scrum and SOA principles are not in conflict with each other because they both are adaptive approaches [2]. Meanwhile SOA with a traditional approach, for example, waterfall model, becomes predictive, as predictive methods completely depend on the requirement analysis and careful planning at the beginning of the cycle. Any change that is to be included will go through a strict change control management and prioritization.

The agile model uses an adaptive approach where there is no detailed planning and only clear future tasks are those related to the characteristics that must be developed. The team adapts to dynamic changes in the product requirements. The product is frequently tested, minimizing the risk of major faults in the future. Interaction with the clients is the strong point of agile methodology and open communication and minimal documentation are typical characteristics of the agile development environment. Teams collaborate closely and are often located in the same geographical space. Therefore, an agile model is better suited for the rapid development of adaptive services in SOA projects development. Application development through Scrum without a clear and strong idea of the aims of the organization will be useless. SOA without a clear image of how exactly to design and build using Scrum process model rules is a waste of resource and time. Also using SOA, as a pervasive strategy, for developing software application is increasing, since it focuses on the ability to respond to changes. Since adapting to changes is the indispensable

concept of SOA development as well as agile development, it seems that using agile methodology is a natural fit to develop SOA applications.

Hence, Scrum and SOA are about agility that can be used together by applying a number of rules and principles, which are not in conflict with each other. This way they maintain each other in balance.

5.2. Scrum and SOA Commonalities. As already discussed, SOA is an architectural approach that focuses on the fact that business organizations must be intelligent and could respond to rapid changes in business. By developing services, one can reach closer to the indefinable goal of software reuse. In SOA approach, different teams build individual services and then these services are integrated in an application. [49, 52]. Meanwhile Scrum is an agile process model for application development, which stresses on responding to changes [8]. By introducing both technical and nontechnical practices, teams are able to support businesses to become agile. Scrum and SOA approaches are paired by nature and share general objectives. In both cases, change is adaptable, and organizations need to effectively cope with that change. Therefore, Scrum could be a choice when building SOA based applications.

5.3. Adversity of SOA and Scrum. Although Scrum and SOA are compatible, still there are some major distinctions among the two approaches, which should be understood and handled while working with SOA and Scrum together in a

TABLE 2: Scrum and SOA common metrics.

	Scrum	SOA	Commonalities
1	Completed stories versus planned stories	Newly created & used as a percentage of total services	The main purpose of these two metrics is to measure the ratio and percentage of completed work (services) developed in one sprint.
2	Team velocity	Development time & average development time to develop a service	These three metrics measure the team progress in terms of sprint and time required for a service which is to be completed in a particular sprint within time. So these metrics could be combined into team velocity metric which will be considered as a metric for SOA and Scrum integration.
3	Quality delivered to customer	Service quality assurance	The aim of these metrics is to measure the service quality when applying the Scrum development process model. The quality is a common feature for both metrics which can be combined to make a metric for SOA and Scrum integrations measurements.
4	Team enthusiasm	Violation of architecture policies	When the team members are happy and satisfied and work in comfortable environment, then they will communicate with each other collaboratively and in a friendly way. They will have full attention and focus on product development through this product quality will remain standard. Also they will willingly follow the preplanned architecture policies and rules. When the Scrum team is happy and in restful environment, then they can develop a large number of services of high quality in small amount of time. So we can say that the “team enthusiasm & communication” metrics of Scrum and “violations of architecture policies” & “average time to service development” metrics are dependent on each other; these can have an effect on project when these are not concentrated. These metrics are used to measure the behavior of how they follow rules and policies during development environment.
5	Team communication	Average time to service development	
6	Retrospective process improvement	Service accessibility & usability	These two metrics can be integrated together to represent a common metric for both Scrum and SOA. Because the retrospective meeting is held in the last of all practices of Scrum in which the overall activities could be revived, when the services are developed in a sprint, a review meeting will be arranged in which we can test the developed service functionality and usability of how to access the service and how it works.
7	Technical debt management	Reduction in the project and maintenance expense	The main purpose of these two metrics is to reduce the product development cost through best management and utilization of resources and team member's skills. These two metrics can be integrated in one common metric for combined use of Scrum and SOA approaches.

project [53]. One of the leading causes is that the two approaches have different origins and diverse directions. Scrum is traditionally small-project-based, although with process improvement and experience of practitioners have gain knowledge and learned experience to adapt the rules of the Scrum manifesto can also be applied to large software development projects. SOA is a top-down and divide-and-conquer approach to applications development. The “divide” part naturally results in low communication between teams. There are three main areas where Scrum and SOA clash with each other.

- (1) SOA encourages that architecture be designed upfront, while in Scrum community big design upfront (BDUF) is considered as an antipattern.
- (2) SOA encourages teams to split along functional lines, while Scrum encourages cross-functional teams.
- (3) SOA does not have any formal feedback and communication among development teams, while Scrum

is focused on frequent feedback at both a technical level and personal level.

When using SOA in a large setup, it may be complicated to effect any change [54]. There may also be other issues beyond software agility and cost, for example, lack of team communication and discoverability of services. So instead we should strive for closer cooperation and openness among teams. Each team will make available an easy modifiable version of its service for internal test. The service team will separate test data from service, so it survives even if the code is messed up. The service team will provide clean and simple documentation on how to add new calls to the service stub.

6. Conclusion

Service-oriented architecture (SOA) allows reusing the functionality of existing systems rather than building again from scratch. This feature of reusability in the SOA based

applications maximizes economic benefits for organizations. Scrum process model, on the other hand, tends to focus on iterations and client suggestions to improve performance and allow for the predictability of varying requirements. The study analyzes a basis for identifying commonalities and compatibilities in Scrum and SOA process to achieve maximum benefits of the organized Scrum management process for SOA based applications development. The study establishes that most of the Scrum and SOA principles are not in conflict with each other. Both Scrum and SOA are about agility which can be applied using rules and principles that do not clash with each other. To confirm this compatibility, the performance of an integrated Scrum and SOA development environment can be tested through formal KPIs based on the individual Scrum and SOA metrics commonly used by Scrum and SOA practitioners in the software development industry. These KPIs will provide a formal approach to measure agility, complexity, efficiency, and value of Scrum and SOA for those teams who want to use the Scrum and SOA in an integrated environment. Although the output of the study is complete, the researchers identify some limitations which should be considered before applying the results for an industrial Scrum and SOA integrated project.

7. Future Work

The following are some of the important points for future work based on the outcome of this research:

- (1) To find out what will be the impact of Scrum and SOA integration on the project performance if used in a distributed environment.
- (2) To define a formal process for the evaluation of identified metrics and KPIs in nondistributed and distributed Scrum and SOA integrated projects.

Data Availability

No data are available.

Conflicts of Interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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